

Adaptive localization by observation assignment for the modified-Cholesky ensemble Kalman filter

Working draft

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Draft status. The selection rule of Section 3 (the cost J) is superseded: it selects the least correlated candidate (see `nota_criterio.tex`). The method now reads each component's radius from the partial correlations of a probe modified-Cholesky precision (`qgloc/assignment.py`, `assign_partial`); the analysis of Section 4 and the benchmark design are unchanged. Sections 3 and 5 are to be rewritten from the tables `tables_paper.tex` once EXP-02 has run.

Abstract

Localization in ensemble data assimilation is normally parameterized by a radius of influence, tuned once against a known truth and then held fixed. We replace the radius by a different object: each model component is updated by exactly one observation, chosen from the few that its neighbourhood reaches, and the radius is whatever distance that choice implies. The choice is made by minimizing a variational cost evaluated at a scalar local analysis, computed from ensemble anomalies and observations alone, with no reference trajectory and nothing withheld. Because the cost decomposes over components, the optimal assignment is obtained exactly, in closed form, at a cost of microseconds per evaluation. The resulting radius field adapts to the local observation density without any parameter controlling it, and the observation selected is very often not the nearest one: on a quasi-geostrophic testbed the second, third and fourth candidates are chosen about as often as the first, because what decides the choice is the ensemble correlation and not the distance. The assignment determines the predecessor sets of a modified-Cholesky precision matrix, which is assembled once per cycle and used for a single global analysis.

1 The parameter nobody can pin down

The analysis step of an ensemble Kalman filter combines a forecast ensemble with observations. With an ensemble far smaller than the state dimension, the sample covariance is rank deficient and carries spurious correlations between distant variables; localization removes them. Every localization scheme depends on a radius of influence, and the radius is normally chosen by sweeping it against a known truth in a synthetic experiment and then freezing it.

Attempts to estimate the radius instead run into a difficulty that is not a matter of tuning. A radius per model component is a free parameter with nothing in the data to pin it down. In experiments preceding this work, on a system of 40 components, an oracle with full access to the truth reduced the analysis error by 59% on the cycle it optimized, and the same radius field made the error 22% *worse* on the following cycle, performing worse than a random shuffle of its own values. What such a search finds is the cycle's noise.

Grouping components reduces the count but introduces a grouping that must itself be justified, and the radius within a group remains a free number.

2 Assignment instead of radius

We propose a different parameterization. Let the model state have components $i = 1, \dots, n$ and let there be m observations at known locations. For each component, grow a neighbourhood

until it contains κ observations; call them the *candidates* of i and write $\mathcal{C}(i)$ for the set. The component is then updated by exactly one member of $\mathcal{C}(i)$, or by none at all.

The decision variable is the assignment

$$a : \{1, \dots, n\} \longrightarrow \mathcal{C}(i) \cup \{\emptyset\}, \quad (1)$$

and the radius of component i is not chosen: it is the distance to $a(i)$.

Three properties follow immediately.

The radius adapts to the observation density with no parameter. Where observations are dense the neighbourhood stops growing early; where they are sparse it grows further. On the testbed below, a regular lattice of spacing four gives a mean radius of 2.7 and a maximum of 4, while a random network of the same overall density gives a mean of 1.7 and a maximum of 5 — the second number is larger precisely because a random network leaves empty regions. No setting controls this.

The nearest observation is not automatically the right one. What determines how much an observation tells us about a component is the ensemble correlation between them, not the distance. In a flow with structure, two points aligned along a current can be far better correlated than two adjacent points across a front. This is why the neighbourhood keeps growing past the first observation: the choice is made on the cost, and the cost sees the correlation.

Abstaining is a legal choice. If none of the candidates correlates with the component, updating it from any of them injects noise, which is what localization exists to prevent. Leaving the component at its background value is one of the options, and the cost selects it on its own.

3 The cost

For component i assigned to observation j , the local analysis is the scalar Kalman update

$$x_i^a = x_i^b + \frac{\sigma_{ij}}{\sigma_{jj} + r_j} (y_j - x_j^b), \quad (2)$$

with σ_{ij} the ensemble covariance between component i and the observed site of j , σ_{jj} the ensemble variance there, and r_j the observation error variance. The cost is the variational one evaluated at that analysis,

$$J_i(a) = \underbrace{\frac{(x_i^a - x_i^b)^2}{\sigma_{ii}}}_{\text{departure from the background}} + \underbrace{\frac{(y_j - x_j^a)^2}{r_j}}_{\text{residual}}, \quad j = a(i), \quad (3)$$

and $J_i = (y_j - x_j^b)^2/r_j$ when i abstains, since the background is kept and the departure term vanishes. The global score is

$$J(a) = \frac{1}{n} \sum_{i=1}^n J_i(a). \quad (4)$$

No reference trajectory enters (3), and no observation is withheld. Everything is a dot product over the N ensemble members. An assignment that moves the state a long way without justification pays in the first term; one that leaves a residual it could have removed pays in the second.

Proposition 1 (Exact minimization). *J decomposes over components, so $\arg \min_a J(a)$ is obtained by minimizing each J_i independently over $\mathcal{C}(i) \cup \{\emptyset\}$.*

Algorithm 1 One assimilation cycle

- 1: propagate the ensemble; form anomalies $\Delta\mathbf{X}$
 - 2: **for each component** grow its neighbourhood until it holds κ observations
 - 3: precompute σ_{ii} , σ_{jj} and σ_{ij} for every candidate pair
 - 4: **for each component** take $a(i) = \arg \min J_i$ over $\mathcal{C}(i) \cup \{\emptyset\}$
 - 5: read off the radius of every component from a
 - 6: assemble $\hat{\mathbf{B}}^{-1}$ once and solve for the analysis
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Remark 1. *This is worth stating plainly because it removes a whole apparatus. There is no search here: the optimal assignment is a per-component $\arg \min$ over $\kappa+1$ precomputed numbers, and it is obtained in one pass over the cost table. The decomposition is a property of the cost (4), not of the implementation.*

4 From assignment to analysis

The assignment determines a radius per component, and therefore the predecessor sets of the modified-Cholesky estimator [1, 2]. For each component i with predecessors $P(i)$ given by its radius,

$$\beta_i = (\mathbf{X}_P^\top \mathbf{X}_P + a_i \mathbf{I})^{-1} \mathbf{X}_P^\top \Delta \mathbf{X}_i, \quad a_i = \alpha \frac{\text{tr}(\mathbf{X}_P^\top \mathbf{X}_P)}{|P(i)|}, \quad \mathbf{L}_{ii} = 1, \quad \mathbf{L}_{i,P(i)} = -\beta_i^\top, \quad (5)$$

with $\mathbf{D}_{ii}$ the inverse residual variance, and $\hat{\mathbf{B}}^{-1} = \mathbf{L}^\top \mathbf{D} \mathbf{L}$. The ridge is a fraction α of the mean eigenvalue of each row's own Gram matrix rather than an absolute constant. This matters here more than in the fixed-radius estimator: the assignment produces rows with anywhere from zero to over a hundred predecessors in the same cycle, and an absolute penalty would regularize the short rows hard and the wide rows not at all. The relative form gives α the same meaning in every row, every cycle and every configuration compared below; $\alpha = 0.3$ throughout. The analysis solves

$$(\hat{\mathbf{B}}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}) \delta \mathbf{x} = \mathbf{H}^\top \mathbf{R}^{-1} (\mathbf{y} - \mathbf{H} \mathbf{x}^b). \quad (6)$$

The precision is positive definite for any assignment, because it retains the form $\mathbf{L}^\top \mathbf{D} \mathbf{L}$; truncating an already-formed precision does not preserve this, and in a companion measurement produced an indefinite matrix in 44 of 44 cases tested.

5 Numerical experiments

The testbed is the 1.5-layer quasi-geostrophic model of Sakov and Oke [3],

$$q_t = -\psi_x - \varepsilon J(\psi, q) - A \Delta^3 \psi + 2\pi \sin(2\pi y), \quad q = \Delta \psi - F \psi, \quad (7)$$

on the unit square with $F = 1600$, $\varepsilon = 10^{-5}$ and Dirichlet boundaries, at 49×49 per field. The integration is carried on q and ψ is recovered from it at every step, so ψ is diagnostic and only q is estimated. q vanishes on the Dirichlet boundary for every state, so the boundary components are removed from the assimilation altogether, as in [3]: they are not observed, not candidates, not predecessors, and not in the error. The estimated state is the 47×47 interior, $n = 2209$. Ensemble members and the truth are drawn from snapshots of a long run, following [3]. The dissipation is ten times the model default, which is what gives a stationary regime: at the default the norm of q grows by a factor of fourteen between $t = 500$ and $t = 8000$ and there is no climatology to sample.

`results/figures_paper/radius_fields_lattice_enkf-mc.pdf` — run `make paper`

Figure 1: The radius field of each rule inside the cycled EnKF-MC on the lattice network, on one colormap, and the uniform field of the same mean as the greedy one. Crosses mark observations the greedy assignment does not use.

`tessellation_lattice_enkf-mc.pdf`

Figure 2: Which observation updates each component: the nearest-observation tessellation against the optimal assignment, and the components where the two differ.

5.1 What the assignment does on a forecast

Everything in this subsection is computed from ensemble anomalies and observations before any analysis is performed, so it does not depend on which filter follows. Table ?? gives, for both observation networks and for a climatological ensemble as well as one the filter has produced, the cost of each rule and the structure of the precision the greedy field induces against a uniform radius of the same mean. Three things are stable across seeds, networks and cycles. The cost separates the rules by orders of magnitude, and the optimal assignment improves on the nearest observation by an order of magnitude. Roughly three quarters of the components with a choice do not take their nearest observation: the ensemble correlation, not the distance, decides. And a third of the observations update no component at all; the ensemble variance at those sites is one to two orders of magnitude below the variance at the sites that are used, so the assignment is discarding observations where the Kalman gain would be near zero anyway (Fig. 3).

5.2 The cycled benchmark

Four rules for the radius field are compared inside two filters: the modified-Cholesky EnKF, where the field defines the predecessor sets of $\hat{\mathbf{B}}^{-1}$, and the LETKF, where the same field defines the local domain of each component. The uniform radius $r \in \{1, 2, 3, 4, 6, 8\}$ is the standard scheme and the baseline; greedy is the method; nearest and random are the two controls, one showing what distance alone achieves and the other that a spread of radii is not by itself the point. Both filters run on two networks of equal count: a lattice of spacing two, which shifts by one grid point per cycle, and a random network redrawn every cycle. $N = 40, 60$ cycles with the first 15 discarded, observations every 20 time units with 5% noise, inflation 1.15, $\kappa = 4$, three seeds. In the EnKF-MC a component that abstains has a diagonal precision row; in the LETKF it keeps its background. Divergence — a forecast the model cannot integrate — is recorded per run and counted, not averaged.

Table ?? reports the post-burn-in analysis error of q over the interior; Table ?? what the assignment did inside each filter; Table ?? the cost per cycle; Table ?? the same comparison at $N = 20$, at $\alpha = 0.1$, and on a lattice of spacing three, so that neither the ensemble size, the regularization nor the density is left as a question. Figure 4 places the rules against the uniform-radius sweep.

`paper/tables_paper.tex` is written by `figures/make_tables.py` after `make paper`; until then the tables referenced above are absent.

6 Discussion

The contribution is a reparameterization. The radius stops being a free number and becomes a consequence of a choice among observations that are actually present, which is what makes it estimable: a choice among $\kappa + 1$ alternatives is constrained in a way that a positive real number is not.

discarded_obs_lattice_enkf-mc.png

Figure 3: Ensemble spread of q along the run with the observations that update nothing marked, and the distribution of ensemble variance at used and unused observation sites over all cycles.

rmse_vs_radius_main.pdf

Figure 4: Post-burn-in analysis RMSE against the uniform radius (points, median and range over seeds), with each assignment rule as a horizontal band placed at its mean radius (diamond), for both filters and both networks.

Because the cost decomposes, the optimum is exact and costs one pass over a table of $n(\kappa+1)$ numbers. The whole expense of the method is in precomputing that table from ensemble anomalies once per cycle, and in assembling the precision from the assignment that results. Nothing is tuned: κ sets how far a neighbourhood grows before it stops, and every other quantity — the radius of each component, which observations are used and which are not — is read off the assignment.

References

- [1] P. J. Bickel and E. Levina. Regularized estimation of large covariance matrices. *The Annals of Statistics*, 36(1):199–227, 2008.
- [2] E. D. Nino-Ruiz, A. Sandu, and X. Deng. An ensemble Kalman filter implementation based on modified Cholesky decomposition for inverse covariance matrix estimation. *SIAM Journal on Scientific Computing*, 40(2):A867–A886, 2018.
- [3] P. Sakov and P. R. Oke. A deterministic formulation of the ensemble Kalman filter: an alternative to ensemble square root filters. *Tellus A*, 60(2):361–371, 2008.